

■ SQL Revision Notes: COUNT, DISTINCT, & LIMIT

1. Aggregating Data with COUNT

The COUNT function is an aggregate function used to return the number of rows that match a specified criterion. It tells you how many records there are.

	Explanation	Example
	Counts all rows, including those with NULL values. Best for counting total records.	SELECT COUNT(*) FROM Customers
me)	Counts non-NULL values in a specific column.	SELECT COUNT(Email) FROM Employees
Column)	Counts the number of unique, non-NULL values in a column.	SELECT COUNT(DISTINCT JobTitle) FROM Employees

Using COUNT with WHERE

SQL

```
SELECT COUNT(*)
FROM Products
WHERE Category = 'Electronics';
-- Result: Returns the number of products in the 'Electronics' category.
```

2. Refining Data with DISTINCT

The DISTINCT keyword is used to eliminate duplicate values from the result set, showing only the unique values.

	Explanation	Example
DISTINCT	Returns a list of unique values in the specified column.	SELECT DISTINCT City FROM Addresses;
CT with Multiple Columns	Returns unique combinations of the listed columns.	SELECT DISTINCT FirstName, LastName FROM Employees

DISTINCT vs. COUNT(DISTINCT)

SELECT DISTINCT City gives you a list of unique city names (the actual data).
SELECT COUNT(DISTINCT City) gives you a single number (the total count of unique city names).

3. Controlling Output with LIMIT

The LIMIT clause is used to restrict the number of rows returned by the query. This is especially useful for testing queries on large databases or getting the top few results.

	Explanation	Example
and LIMIT	Restricts the output to the first N rows.	SELECT * FROM Orders LIMIT 10;
with OFFSET	Skips the first M rows, then returns the next N rows (used for pagination).	SELECT * FROM Products LIMIT 10 OFFSET 5;

Note: LIMIT is a non-standard SQL extension but is supported by most popular databases (MySQL, PostgreSQL, SQLite). In SQL Server/MS Access, you would use TOP.

4. Executing Complex Queries (The Query Flow)

When you combine SELECT, COUNT, DISTINCT, WHERE, and LIMIT, the database processes them in a specific order:

- 1■■ **FROM:** The database identifies the target table.
- 2■■ **WHERE:** Filters the rows based on the criteria.
- 3■■ **COUNT/DISTINCT:** Operates on the remaining rows (counting or finding unique values).
- 4■■ **SELECT:** Specifies the final columns to be displayed.
- 5■■ **LIMIT:** Truncates the final result set to the specified number of rows.

Combined Example

Find the number of unique customers from California, but only show the count from the top 50,000 records checked.

SQL

```
SELECT COUNT(DISTINCT CustomerID)
FROM Customers
WHERE State = 'CA'
LIMIT 50000;
```

Keep practicing these combinations—they are critical to becoming a proficient SQL user! ■